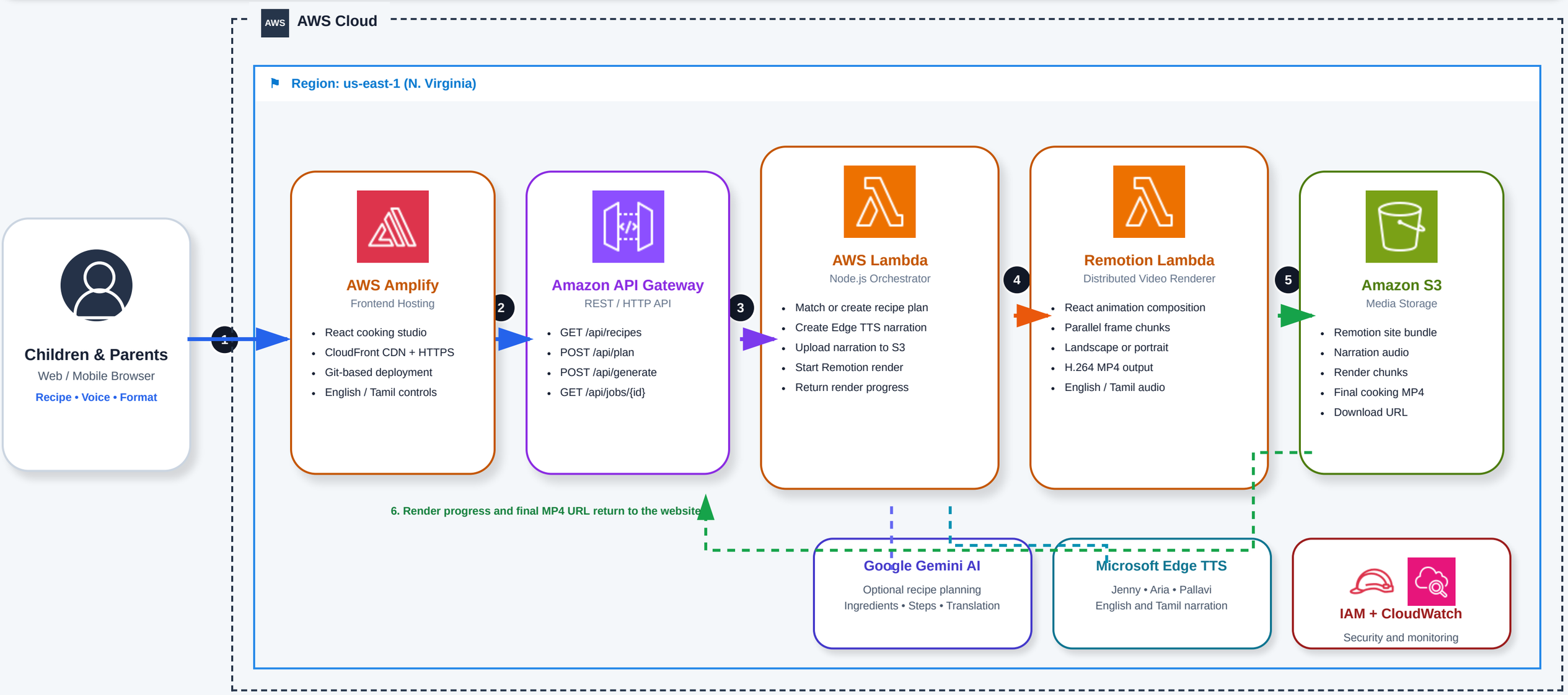
LifeMotion Cooking Studio — AWS Architecture

AI-powered cartoon recipe video platform | Region: us-east-1 (N. Virginia) | Remotion 4.0

AWS Architecture 2026



Tier 1: User Access & API

- **User Access:** Children or parents open LifeMotion through AWS Amplify, CloudFront and HTTPS.
- **API Routing:** API Gateway receives recipe planning, generation and render-progress requests.

Tier 2: AI, Narration & Video Rendering

- **Recipe + Voice:** Lambda prepares the recipe plan, calls optional Gemini and creates English or Tamil narration.
- **Remotion Rendering:** Lambda workers render Chef Maya in parallel and combine the frames into an H.264 MP4.

Tier 3: Storage, Delivery & Operations

- **Media Delivery:** S3 stores narration, render chunks and the final MP4, then returns the download URL.
- **Security + Monitoring:** IAM limits service permissions and CloudWatch records errors, duration and usage.